Chapter 5

Getting Started with Microsoft Copilot

IN THIS CHAPTER

- **Knowing about Copilot's capabilities and benefits**
- **Accessing Copilot**
- **Experimenting with basic prompts**

Microsoft Copilot is an umbrella brand name for all of Microsoft's AI-powered chatbots. Chatbots such as Copilot and similar products from OpenAI, Google, Apple, and many others have the potential to change the way people get work done. At their best, AI chatbots can enhance productivity, learning, and creativity. At their worst, they can produce low-quality text and images, confidently answer questions with fabricated data, and displace human jobs.

In this chapter, you learn some of the ways that you can access Microsoft Copilot, and you get an overview of its history, capabilities, benefits, and limitations.

Defining Copilot

In 2019, Microsoft invested in the then-tiny AI startup called OpenAI. Microsoft provided billions of dollars, and OpenAI ran its systems on Microsoft's computers. In 2021, Microsoft exclusively licensed OpenAI's GPT-3 model, which was used to create OpenAI Codex. OpenAI Codex was subsequently used by GitHub — a subsidiary of Microsoft that provides

tools and hosting for computer programmers — to create a computer programming assistant called GitHub Copilot, shown in Figure 5-1.

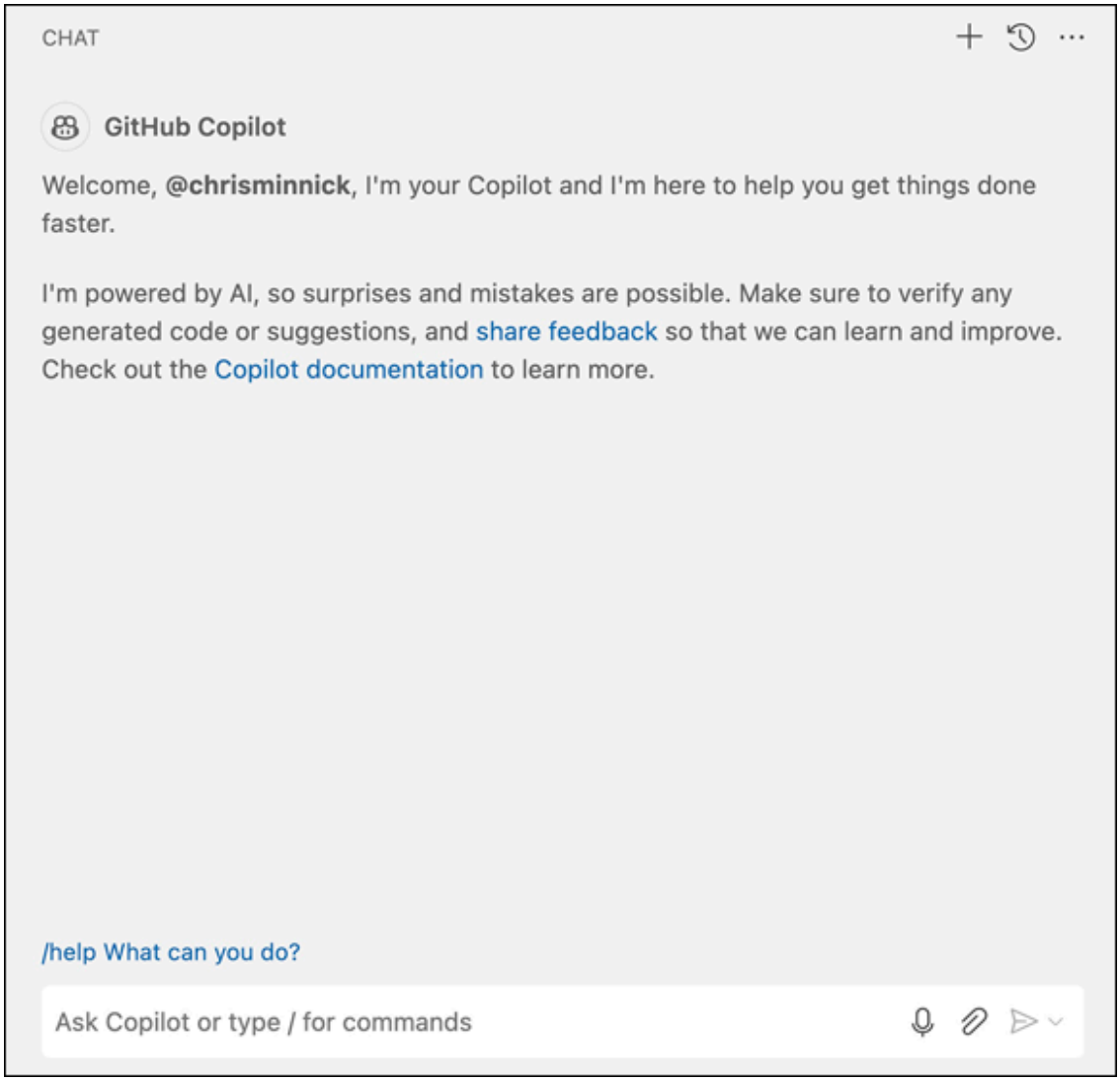


FIGURE 5-1: GitHub Copilot.



REMEMBER Although GitHub Copilot and Microsoft Copilot are similar in that they both use OpenAI's technology for understanding and generating language, they're two different products. GitHub Copilot is optimized to help with the writing of programming code, and Microsoft Copilot is optimized for chatting with people and generating written and spoken words in human languages.

Several months after GitHub Copilot was rolled out as a plugin for Microsoft's Visual Studio Code editor, OpenAI released the first version of ChatGPT for use by the public. ChatGPT became the fastest-growing con-

sumer internet app of all time — gaining 100 million monthly users in just two months.

With its unprecedented ability to respond to user queries with human-like text, ChatGPT became a cultural sensation and possibly even a threat to the traditional search engines created by Google and Microsoft.

Microsoft responded to ChatGPT by redesigning its Bing search engine. *Bing Chat,* as it was called, was rolled out starting in February 2023 and gained its first 100 million active users within months. The early version of Bing Chat tended to produce false data (also known as *hallucinations*) and troubling responses during chats, including, as reported by Kevin Roose in a *The New York Times* article, acting like a "moody, manic-depressive teenager who has been trapped, against its will, inside a second-rate search engine."



TIP *Hallucinations*, in AI lingo, are defined as incorrect or misleading information generated by AI. They're caused by a variety of factors, including insufficient training, incorrect assumptions, and biases in the data used to train the AI model.

Microsoft clamped down on much of Bing Chat's tendency to go off the rails and rebranded it as Microsoft Copilot. The current homepage for Microsoft Copilot (`https://copilot.microsoft.com`) is shown in Figure 5-2.

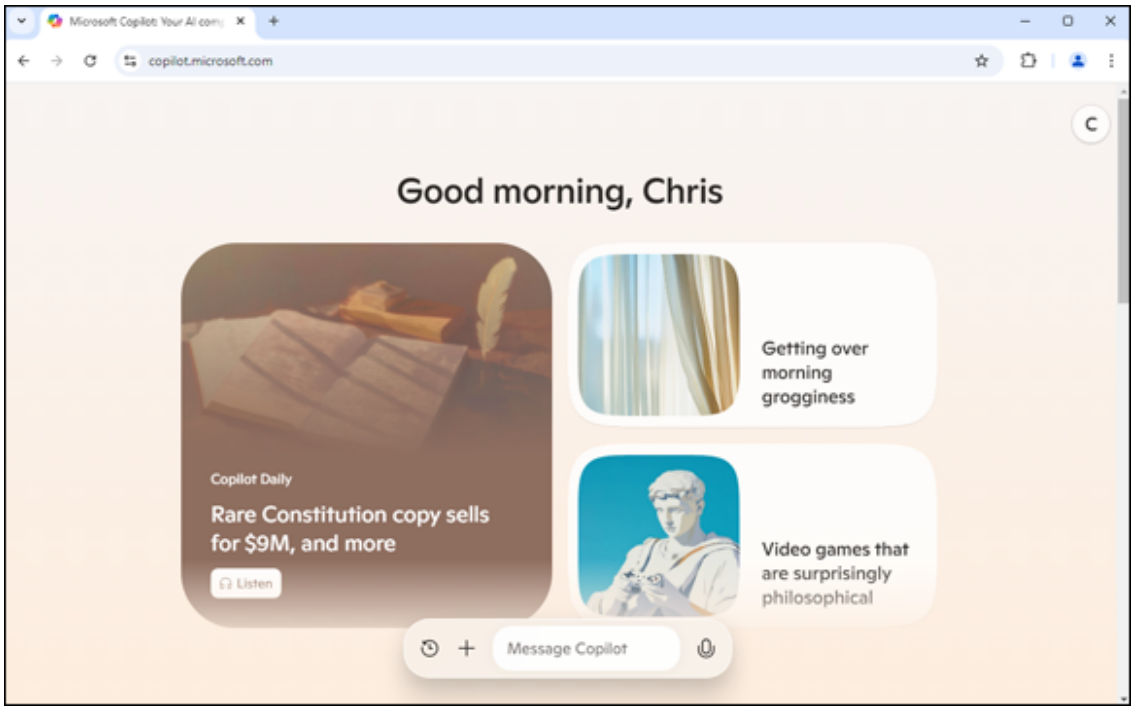


FIGURE 5-2: Microsoft Copilot on the web.

Overview of Microsoft Copilot

Copilot has been integrated into many of Microsoft's products and can be helpful with a wide variety of tasks. In fact, there are so many possible ways to use Copilot that the possibilities can sometimes seem overwhelming. AI chatbots are a fundamentally different way of interacting with computers than most people are used to, so it can be helpful to look at them as if they were traditional computer software and start by talking about the features and what makes Microsoft Copilot different from its competition.

Core functionalities and benefits

The most basic function of any chatbot, whether it's powered by artificial intelligence or not, is to respond to human speech or writing (which is also known as *natural language*) with easy-to-understand text or speech. The quality of a chatbot can be measured by how human-like its responses are.

The current crop of AI chatbots can all generate highly convincing natural language responses to people's questions and requests.

The voice or text input a user of a chatbot gives to the chatbot, whether it's a question ("How tall is Mount Everest?") or an instruction ("Summarize this email.") is called a *prompt*. The primary way for people to interact with chatbots is through *prompting*.

Beyond its core ability to respond to prompts in natural language, Microsoft Copilot has exciting additional capabilities that make it stand out in usefulness, especially when it's integrated into other Microsoft products.

Some of the features of Copilot include the following:

- Performs web searches by using Bing
- Integrates with other AI tools to create original images and music
- Writes original text or rewrites existing text
- Cites the sources of the text it generates
- Personalizes its interactions with you based on previous interactions and documents you work on
- Translates text between different languages
- Supports plugins that expand Copilot's capabilities
- Supports user-created chatbots

Key differentiators from other AI assistants

The main thing that makes Microsoft Copilot more useful than other AI assistants is that it is integrated into Microsoft Windows and other Microsoft programs. This integration gives Copilot the ability to not only generate text and images, but also to control certain aspects of the software it's integrated into. For example,

- When using a non-integrated chatbot, such as ChatGPT, you can ask for text for a PowerPoint slide that you then need to copy into PowerPoint and format manually.
- With Microsoft 365 Copilot, you can ask for PowerPoint slides or an entire presentation, and the Copilot assistant will create the new slides, format them for you, and insert them directly into your presentation. Even better, Copilot can access and use other documents you've created while creating the new slides.



REMEMBER Another key factor that distinguishes Copilot from many other chatbots is that Copilot has access to the data in Microsoft Bing. By augmenting the data it was originally trained on with search results from Bing, Copilot can answer questions about the latest news and other developments, whereas other models have a "cutoff date" beyond which they can only speculate (or hallucinate).

Understanding how Copilot works

Chatbots like Microsoft Copilot and ChatGPT are far superior to their predecessors, such as Office Assistant, aka "Clippy" (shown in Figure 5-3). Microsoft integrated Clippy into Microsoft Office applications from version 97 to 2003, and it proved to be more annoying than helpful in most cases.

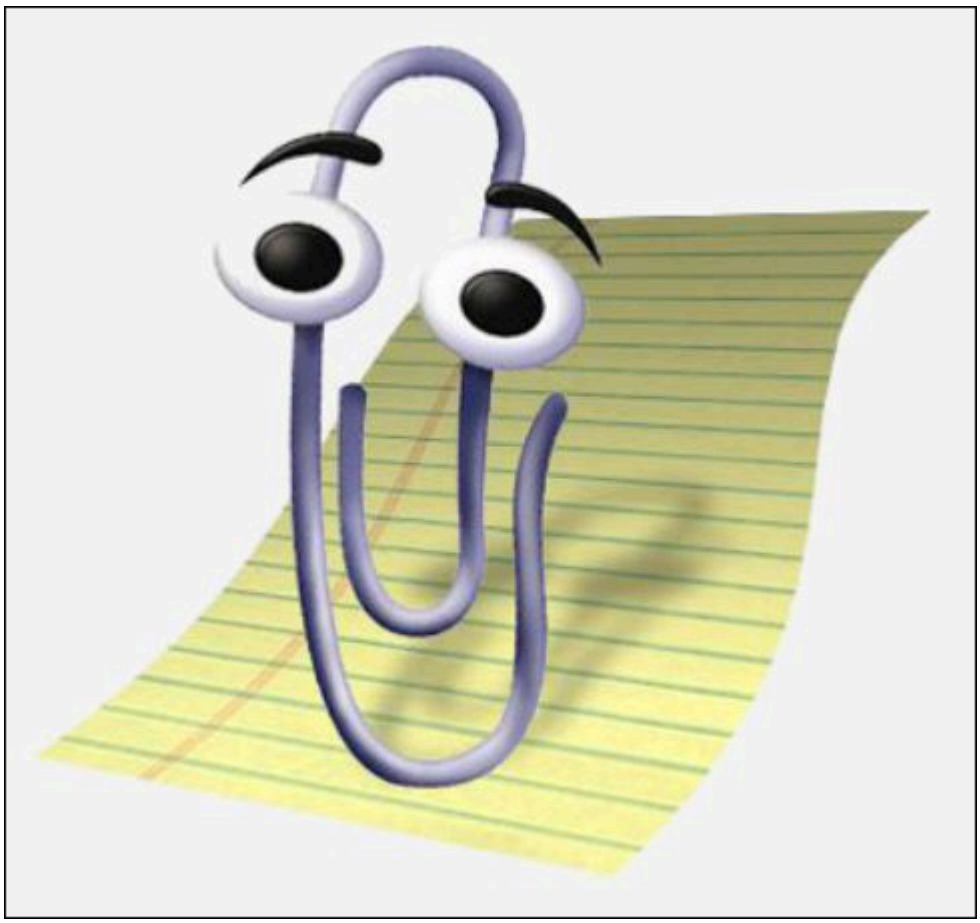


FIGURE 5-3: Original Office Assistant, also known as "Clippy."

The reasons for Clippy's failure have been studied exhaustively, but the crux of it is that Clippy was intrusive and would appear whenever it detected that you were doing something (such as writing a letter) that it was supposed to be able to help with. But then, when you agreed to let Clippy help you, all it could do was reference official Microsoft Office documentation, which wasn't helpful for much of anything.

The two most important factors that contributed to making the latest generation of AI so much better than Clippy (and all subsequent AI assistants) are

- Vastly more data (and computing power) went into their training.
- They take advantage of new AI techniques that allow them to consider context when generating responses.

Learning from all the data

The AI model behind Microsoft Copilot is named *Prometheus* and is OpenAI's technology combined with Bing's search index. The result is that Copilot has learned from and has access to a tremendous amount of data.



Although the relationship between training data size and a model's performance isn't simple, in general, larger models are able to gain a better picture of whatever they're designed to simulate (such as communicating using natural language, in the case of a chatbot).

Context is key

Even more important than simply throwing more data at an AI system is a set of techniques known as *attention*, which Google invented in 2017. In short, what attention techniques do is allow AI models to look at different parts of your input and their own output while figuring out what to say. For example, consider the following sentence:

> *"The bank can guarantee deposits will be safe because it has invested in secure vaults."*

An AI model that uses attention mechanisms will know that "bank" refers to a financial institution rather than the bank of a river because of the other words in the sentence, such as "guarantee," "deposits," "safe," and "vaults." A model with attention also understands that the word "it" in this sentence refers back to "bank."



TIP Because large language models (LLMs, such as CoPilot) can take context into account, providing sufficient context to the model in your prompts has become the single best way to improve the quality of responses you get to your prompts. The section "Basic commands and interactions," later in this chapter, offers a look at a series of human prompts and CoPilot responses.

Integration with Microsoft 365 apps

Microsoft 365 is the family of products and services that includes the productivity programs formerly known as Microsoft Office, as well as the OneDrive cloud storage service, the Microsoft Teams collaboration and conferencing program, the Outlook email and calendar program, and others. Microsoft 365 Copilot is available as an additional subscription.

Subscribing to Microsoft 365 Copilot activates the Copilot chatbot in each application and enables Copilot's built-in actions, which can perform different tasks depending on the application. Some of the features that Microsoft 365 Copilot enables include

- **In Word,** Copilot can suggest different writing styles and formats, rewrite sentences or paragraphs, translate text into other languages, and convert text into tables.
- **In Excel,** Copilot can analyze data to discover trends and insights you may have missed, create charts and graphs, and suggest formulas.
- **In PowerPoint,** Copilot can suggest design ideas, create individual slides, convert Word documents into presentations, add animations, and even write speaker notes.
- **In Teams,** Copilot can take meeting notes, transcribe recordings, summarize discussions, and suggest action items.

- **In Outlook,** Copilot can summarize emails, assist you with writing emails, schedule meetings, and create reminders based on the content of your emails.

If you don't seek out Copilot's help while you're using Microsoft 365, it will remain quietly in the background. This is a welcome change from the overly eager assistant days of Clippy, but it also makes it important for users to educate themselves about what Copilot is capable of helping with.

In Chapters 2, 3, and 4 of Book 3, you experiment with Copilot's integration with Microsoft 365 and start to see its amazing capabilities as well as its sometimes frustrating limitations.

Signing Up for Copilot

Signing up to use Copilot couldn't be easier. I mean that literally, because you don't need any kind of account to try it out. All you need to do is open any web browser and go to `https://copilot.microsoft.com`. The free and logged-out version of Copilot Chat, shown in Figure 5-4, is limited compared to the version you get when you log in with a Microsoft account or buy a subscription to Copilot Pro.

In this section, you learn how to access Copilot, how to sign in to Copilot, and whether you should subscribe to Copilot Pro.

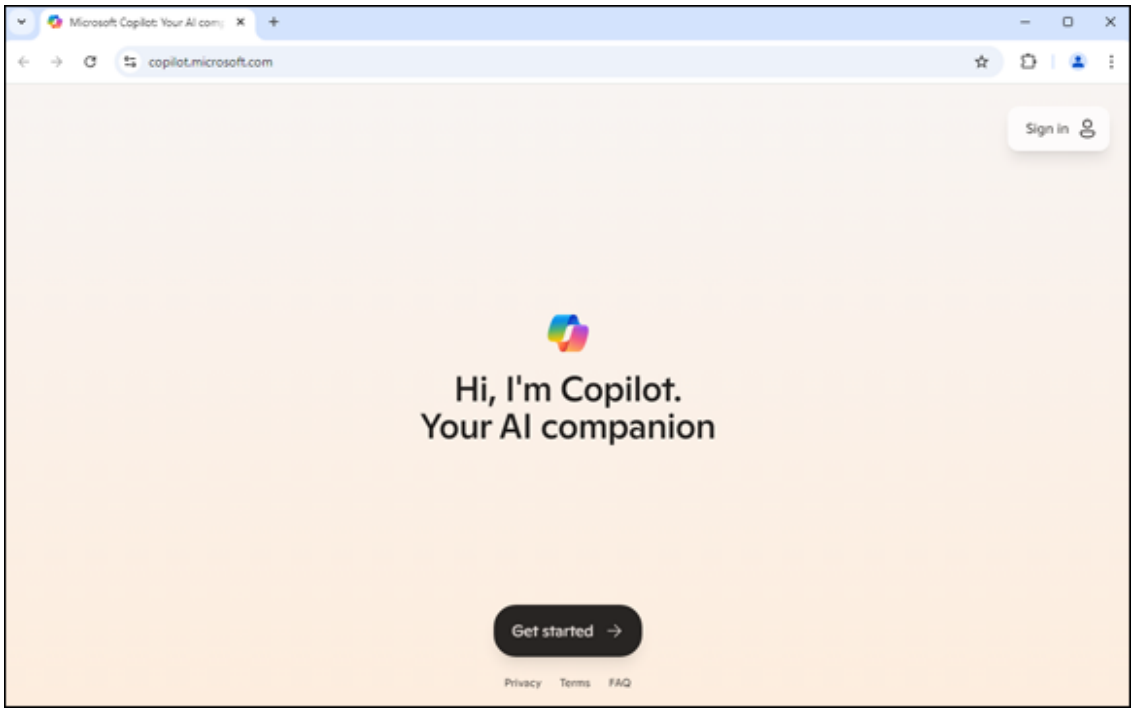


FIGURE 5-4: The free version of Copilot Chat.

Installing Copilot

Because Copilot Chat and Microsoft 365 Copilot both run in your web browser, there's no need to install anything to use either of them. Although the Copilot Chat experience will be nearly the same in any web browser you use, Copilot has integrations with Microsoft's Edge browser that may provide some incentive for users to choose to work with Copilot in Edge. If you have a strong preference for using a browser other than Edge, however, you can use that browser.

Eligibility criteria

Although Bing Chat, as Copilot was previously known, was open to only a limited number of people during its early days, today, Copilot is available for free to anyone with access to a web browser or mobile device.

Subscription plans and pricing

If you use Copilot Chat while not signed in to a Microsoft Account, you're currently limited to just text input and responses. When you try to chat with Copilot by speaking or with images, you'll get a message asking you to log in, as shown in Figure 5-5.

Fortunately, there is an easy fix: Just log in to a free Microsoft account from within Copilot Chat. When you're logged in to Copilot Chat, you'll have access to additional features and settings and to your conversation history.

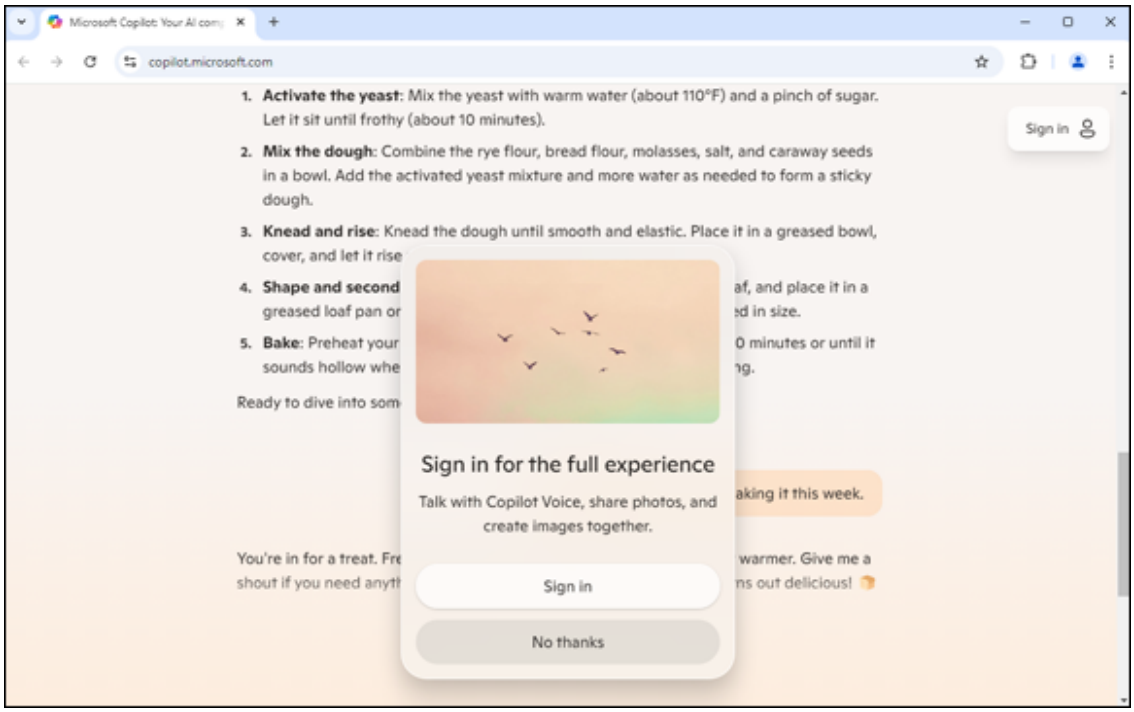


FIGURE 5-5: You have to log in to use certain Copilot features.

Copilot Free

The free and logged-in version of Copilot has its limitations, too, however. These limitations include

- Limited access to the latest and greatest models during times of peak demand
- Limited boost tokens. *Boost tokens* are required for generating images, songs, and other creative content in Copilot.
- No access to Microsoft 365 Copilot
- No custom Copilots. Custom Copilots are specialized AI assistants that users of Copilot Pro can create. You learn more about custom Copilots and how to create them in Book 7, Chapter 4.

Copilot Pro

Subscribing to Copilot Pro currently costs $20/month. The benefits of subscribing include

- Faster performance

- Access to the latest models during peak times
- Copilot access in Office apps
- 100 boost tokens per day, as opposed to the 15 tokens that free users have

If you're already a subscriber to Microsoft 365, you can subscribe to Microsoft 365 Copilot, which gives you the same benefits as Copilot Pro but that is also tuned for business use and integrates with your Microsoft 365 data. Microsoft 365 Copilot is $30/month per user, which is on top of your subscription to Microsoft 365 itself.

Which plan do you need?

If you're a business user and you already have a subscription to Microsoft 365, you should subscribe to Microsoft 365 Copilot.

If you don't have a business license to Microsoft 365 and you plan to do a lot of creative work with Copilot, such as creating images or songs, you should subscribe to Copilot Pro.

If you're just experimenting with Copilot and want to learn what it's capable of and won't be using it every day, the logged-in and free version is more than adequate for your needs.

Step-by-step sign-up process

To sign up for a Microsoft account and get full access to the free version of Copilot, visit `https://copilot.microsoft.com`. After you enter your name and select a voice for Copilot, you'll see a Sign In button in the upper-right corner of the screen, as shown in Figure 5-6.



TIP If you prefer to sign up for and access Copilot using a mobile app, go to `https://www.microsoft.com/en-us/microsoft-copilot/for-individuals/copilot-app` to access a QR code you can scan with your mobile device to go directly to the app download page.

WHY YOUR COPILOT EXPERIENCE DIFFERS

If you see something in Copilot on your computer or at `https://copilot.microsoft.com` that's different from what I show in this book, it's likely because of an upgrade to Copilot that's happened in the time after I wrote the book.

Likewise, if you're not seeing a feature of Copilot that I describe in this book, that also may be because of an upgrade that's happened, or it may be because of an upgrade that hasn't happened for your Microsoft account yet.



FIGURE 5-6: The Sign In button at `copilot.microsoft.com`.

Follow these steps:

1. **Click the Sign In button and select whether you want to sign in with a personal account or a business account.**
   If you'll be signing up for a new account, click Sign In with a Personal Account. If you already have a Microsoft account, you may be logged in automatically at this point. If you don't have an account, you'll see the Sign In page, as shown in Figure 5-7.

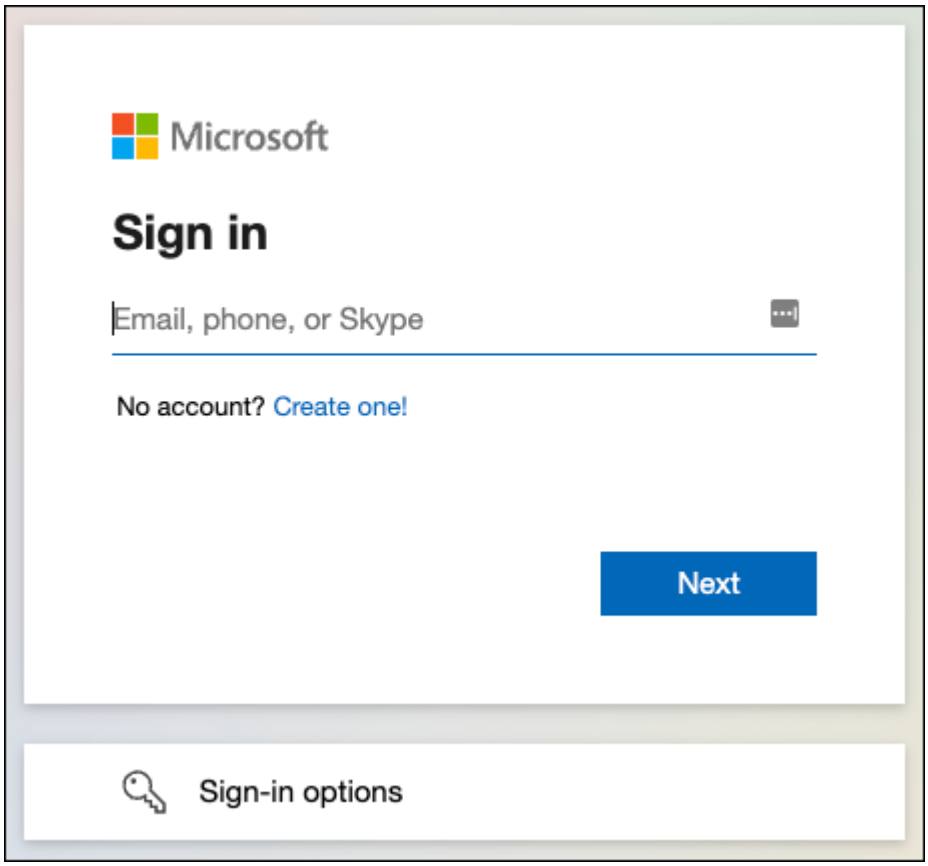


FIGURE 5-7: The Microsoft account sign in page.

2. **To create a new Microsoft account, click the link labeled No Account? Create One!**

   The Create Account screen appears next, as shown in Figure 5-8.

3. **Set up an email address to use with Copilot.**

   You have options:

   1. If you have an email address that you'd like to use as your Microsoft account user ID, you can enter it into this screen.

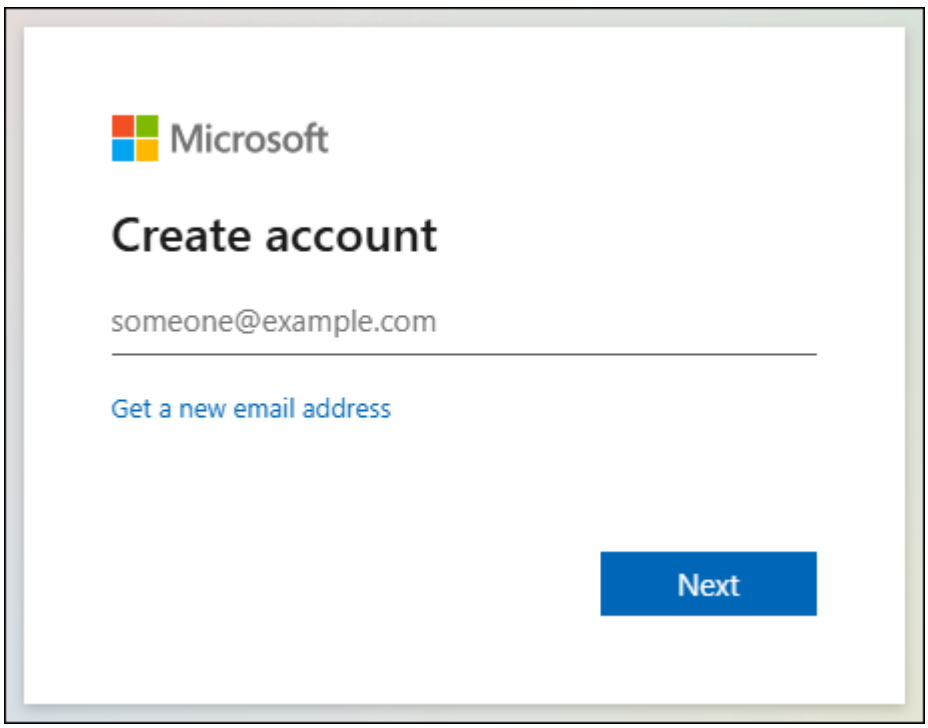


FIGURE 5-8: The Create Account screen.

2. Otherwise, click the Get a New Email Address button to sign up for a free Outlook email address. The first screen of the new email address sign-up page will appear, as shown in Figure 5-9.

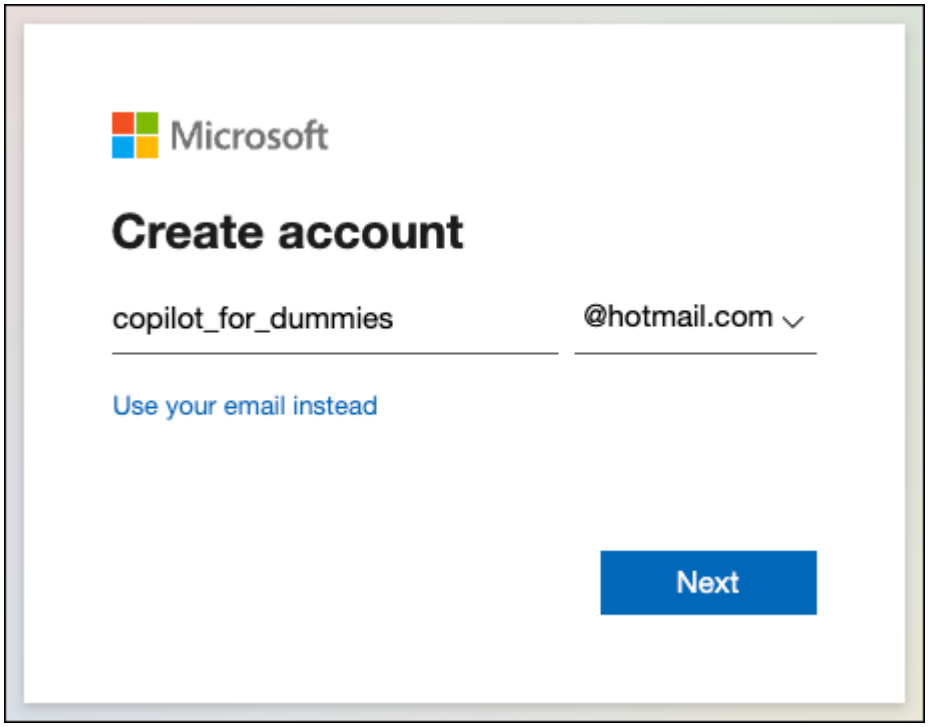


FIGURE 5-9: Selecting a new email address.



TIP When you sign up for a new personal email address through Microsoft, you can choose between an `outlook.com` email address and a `hotmail.com` email address. Which one you choose won't affect how your email address functions in any way, but you might choose one or the other depending on whether your preferred email address is already taken for one of the domains. Or, you might just like the retro vibe of having a `hotmail.com` email address.

4. **On the next screen, select a strong password and check the box next to the question of whether you'd like to receive marketing emails from Microsoft.** Make a note of the password you choose. You'll need this later to log in to your new email account.
5. **Traverse the remaining screens to finish signing up for Copilot**
You'll be asked on subsequent screens for your name and your birthdate. Finally, there may be one last screen that asks you to solve a puzzle to prove that you're human. You've got this!

After you finish signing up for your new email address (which also serves as your new Microsoft account), you'll be returned to the Copilot home-page, and your first name will appear in the upper-right corner where the Sign In button previously was. Congratulations! In the next section, you can start seeing what Copilot can do!

Taking Copilot for a Test Flight

Now that you understand a bit about what Copilot is and how it works, and you've created your account and are logged in, it's time to try it out and see what it can do! The examples in this section use Copilot Chat while logged in to a free account.

Understanding why mine looks different

Because Copilot is an AI model, rather than a simple rules-based program, it will generate different content for the same question each time a question is asked (even if it's only slightly different). So, I show you the results that I get when I enter the same prompts I ask you to enter, but your results will be different. They may be better or worse, but I guarantee that they'll be different.

Prompting and interacting

Let's start with something short and simple. There's not much that's shorter or simpler (at least conceptually) than a haiku, so let's ask Copilot to write some poetry.

Find the input box labeled "Ask Me Anything..." at the bottom of the Copilot Chat window and type the following into it (feel free to customize the topic for one of your favorite things!):

> *Write a haiku about sushi.*

When I tried this, I got the following result:

- *Certainly! Here's a haiku about sushi:*

 *Rice, fish, seaweed hug, Soy sauce whispers on my tongue, Sushi love blooms bright.*

- *Feel free to ask if you'd like more creative content or have any other requests!* 😊

This is okay, I suppose. But, I kind of prefer my poetry with a laugh. So, try giving the following prompt:

> *Write a funny haiku about sushi.*

Here's Copilot's response:

- *Certainly! Here's a playful haiku about sushi:*
   *Rolling down my plate, Avocado slips away, Chasing sushi dreams.*
- *Feel free to ask for more creative content or any other requests!* 😊

Much better! Notice how just the addition of one word (funny) dramatically changed the output and improves it (in my opinion). Let's see if we can get something even better or more descriptive. Here's the next prompt I gave:

> *Write an ironic haiku about sharing sushi.*

With this prompt, I really have no expectations or idea about what I'm looking for. But I do know that my favorite kind of humor is ironic, and one of my least favorite things is having to share sushi (because I want it ALL!). Here's the response I got:

-  *Chopsticks duel begins, Sharing sushi? Not today, Mine, all mine, I say.*
- *Feel free to ask for more creative content or any other requests!* 😊

Again, this is pretty good. My choice of words may be a little more risqué than what Copilot is allowed to use, but I think we've managed to get as close to a decent poem as can be expected.

Feel free to continue experimenting with this prompt or other prompts as long as you like. You can also try out some of the other prompt engineering techniques you learn throughout this book, including in Chapters 2 and 6 of this book.